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INTEGRAL LATCH KEY RING

Abstract

A key ring comprising a ring portion having a first end and a second end where the first end includes a first channel and a catch and the second end includes a second channel and a latch disposed within the second channel of the second end. The latch is moveable between an open position such that the latch is withdrawn into the second channel exposing a gap between the first end and the second end of the ring portion and a closed position such that the latch is moved outward from the second channel to close the gap and engage the first channel and the catch of the first end of the ring portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to: U.S. Provisional Patent Application No. 63/560,358, filed on Mar. 1, 2024, and titled INTEGRAL LATCH KEY RING, (Attorney Docket No. HANDP004.P1) the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] This invention relates to rings for holding keys for ordinary locks, such as for doors or vehicles or any other security lock. Particularly, this invention relates to system and methods for managing the organization and utility of such keys.

2. Description of the Related Art

[0003] Physical security, along with digital security, are essential considerations of modern life. People must take reasonable measures to secure their homes, their workplaces, their vehicles, and any other valuables as much as their digital information. Accordingly, a lock is typically employed to secure any significant property which could be otherwise accessed by someone without permission. Thus, a typical individual will possess a number of keys they must manage that are associated with the various locks they use.

[0004] One very well known prior art ring for keys is a ring effectively made as a spring coiled onto itself. One end of the spring is pulled away from the coil in order for the hole in a key bow to be threaded onto the spring and pushed around the coil until it clears the other end. The key is now completely captured on the ring and free to move around it. Additional keys can be added. Other rings for keys may employ various types of spring loaded latches. One type of spring loaded key ring latch operates similar to a carabiner latch. Another well known type of spring loaded key ring latch employs an over center latch hooked onto a notch across an open gap in the ring.

[0005] In view of the foregoing, there is a need in the art for improved devices and methods for managing and organizing the storage of keys. There is a need for such devices to be simple and compact as well as secure. There is also a need for such devices and methods that operate reliably and efficiently over many uses and at a reduced cost. These and other needs are met by the present invention as detailed hereafter.

SUMMARY OF THE INVENTION

[0006] A key ring comprising a ring portion having a first end and a second end where the first end includes a first channel and a catch and the second end includes a second channel and a latch disposed within the second channel of the second end. The latch is moveable between an open position such that the latch is withdrawn into the second channel exposing a gap between the first end and the second end of the ring portion and a closed position such that the latch is moved outward from the second channel to close the gap and engage the first channel and the catch of the first end of the ring portion. The ring portion and the latch can be wire electro discharge machining (EDM) cut from a metal, such as titanium.

[0007] A typical embodiment of the invention comprises an apparatus having a ring portion having a first end including a first channel and a catch and a second end including a second channel and a gap between the first end and the second end and a latch disposed within the second channel of the second end for opening and closing the gap. The latch is moveable between an open position such that the latch is withdrawn into the second channel opening the gap between the first end and the second end of the ring portion and a closed position such that the latch is moved outward from the second channel to close the gap and engage the first channel and the catch of the first end of the ring portion. In some embodiments, the ring portion can be bounded within a circular envelope and the latch includes a contact area which extends outside the circular envelope. In some

embodiments, the latch can further comprise an open stop latch and the ring portion further comprises an open stop and the open stop latch engages the open stop in the open position. [0008] In further embodiments, the latch can comprise a cantilever spring for engaging the catch of the first end of the ring portion when the latch is in the closed position. The cantilever spring can comprise a latch end turned back on itself. [0009] In still further embodiments, the latch further can comprise a closed stop latch and the ring portion further comprises a closed stop and the closed stop latch engages the closed stop in the closed position. The closed stop latch can comprise a second cantilever spring. Further, the second cantilever spring of the closed stop latch can comprise a slotted back end of the latch. [0010] In other embodiments, the ring portion can further comprise an opening in an outer circumference of the ring portion to the second channel for receiving the latch during assembly. In some embodiments at least the ring portion can comprise titanium, the latch and the ring portion can comprise castings, and/or the ring portion and the latch can be cut using wire electro discharge machining (EDM).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Referring now to the drawings in which like reference numbers represent corresponding parts throughout:

[0012] FIG. 1A illustrates an isometric view of an exemplary key ring embodiment having an integral latch in the closed position;

[0013] FIG. 1B illustrates a side view of an exemplary key ring embodiment having an integral latch in the closed position;

[0014] FIG. 1C illustrates a lower isometric view of an exemplary key ring embodiment having an integral latch in the closed position;

[0015] FIG. 1D illustrates a side view and relative top, front, bottom and back views of an exemplary key ring embodiment having an integral latch in the closed position;

[0016] FIG. 1E illustrates a cutaway view of an exemplary key ring embodiment having an integral latch in the closed position;

[0017] FIG. 2A illustrates an isometric view of an exemplary key ring embodiment having an integral latch in the open position;

[0018] FIG. 2B illustrates a side view of an exemplary key ring embodiment having an integral latch in the open position;

[0019] FIG. 2C illustrates a lower isometric view of an exemplary key ring embodiment having an integral latch in the open position; and

[0020] FIG. 2D illustrates a cutaway view of an exemplary key ring embodiment having an integral latch in the open position.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

1. Overview

[0021] As previously mentioned, embodiments of the invention comprise a key ring having a ring portion having a first end and a second end where the first end includes a first channel and a catch and the second end includes a second channel and a latch disposed within the second channel of the second end. The latch is moveable between an open position such that the latch is withdrawn into the second channel exposing a gap between the first end and the second end of the ring portion and a closed position such that the latch is moved outward from the second channel to close the gap and engage the first channel and the catch of the first end of the ring portion. The ring portion and the latch can be wire electro discharge machining (EDM) cut from a metal, such as titanium. [0022] In one example, the catch can be a hook affixed to the second end extending from the

second end and turning back toward and parallel with the second end and having a tip extending into an enclosed area of the hook. The catch and the rigid loop can be oriented either in or transverse to the plane of the ring portion.

2. Exemplary Integral Latch Key Ring

[0023] FIG. 1A illustrates an isometric view of an exemplary key ring **100** embodiment having an integral latch **104** in the closed position. FIG. 1B illustrates a side view and FIG. 1C illustrates a lower isometric view of the exemplary key ring **100** embodiment having the integral latch in the closed position. In use, the latch **104** is moveable between a closed position and an open position. The latch **104** is disposed within a ring portion **102** and can be readily opened and closed by a simple movement. In a typical embodiment of the device the ring portion **102** comprises a circular shape, i.e. generally fits within a circular envelope, but with a gap **108** between the first end **124** and the second end **126**. The ring portion **102** has a first end **124** which includes a first channel **122** and a catch **120** and a second end **126** which includes a second channel **112**. The latch **104** is disposed within the second channel **112** of the second end **126** for opening and closing the gap **108**. As shown in FIGS. 1A-1E the latch **104** in the closed position spans the gap **108** between the first end **124** and the second end **126** of the ring portion **102**.

[0024] FIG. 1D illustrates a central side view and relative top, front, bottom and back views of the exemplary key ring **100** embodiment having the integral latch **104** in the closed position. Here also, the gap **108** between the ends **124**, **126** of the ring portion **102** is spanned and closed by the latch **104**. In the closed position, the key ring **100** can be used to secure items (e.g. keys) together which have a loop that can be passed through the gap **108** onto the ring portion **102** prior to moving the latch **104** into the closed position.

[0025] FIG. 1E illustrates a cutaway view of an exemplary key ring **100** embodiment having an integral latch **104** in the closed position. This view shows the functional relationship between the integral latch **104** and the ring portion **102** in the closed position. The latch **104** is engaged into the first channel **122** of the first end **124** of the ring portion **102**. The first end **124** of the ring portion **102** also includes a catch **120** within the end of the first channel **122**. The latch **104** can also employ a cantilever spring **106** for engaging the catch **120** of the first end **124** of the ring portion **102** when the latch **104** is in the closed position. The cantilever spring **106** exhibits a small force radially from the center of the ring portion to engage the catch **120**. This feature prevents the latch **104** from accidentally slipping out of the closed position and opening the gap **108**. To release the latch **104**, a small amount of pressure is applied by the user against the cantilever spring **106** to disengage the spring **106** from the catch **120** while moving the latch **104** by force applied to the contact area **110** away from the gap **108**. Although any suitable spring device can be employed as will be understood by those skilled in the art, the cantilever spring **106** can be formed by the latch end turned back on itself as shown.

[0026] A close stop latch **118** at the end of the latch **104** can also be employed to engage the close stop **114** of the ring portion **102** when the latch **104** is in the closed position. The close stop latch **118** can comprise a second cantilever spring. The cantilever spring of close stop latch **118** exhibits a small force radially from the center of the ring portion to engage the close stop **114**. Moreover, this second cantilever spring of the close stop latch **118** can comprise a slotted back end of the latch **104**. As shown, the slot **132** of the latch **104** creates the long cantilever of the close stop latch **118** which is opposed by an inner radial surface **134** of the second channel **112** of the ring portion **102**. In the closed position, the spring **106** engages the catch **120** and prevents movement of the latch **104** towards the open position. In addition, at the opposite end of the latch **104** the close stop latch **118** engages the close stop **114** of the ring portion **102** and prevents further movement of the latch **104** toward the closed position. Accordingly, the latch **104** is trapped from movement in either direction when secured in the closed position.

[0027] FIG. 2A illustrates an isometric view of an exemplary key ring **100** embodiment having an integral latch **104** in the open position. FIG. 2B illustrates a side view and FIG. 2C illustrates a

lower isometric view of the exemplary key ring **100** embodiment having an integral latch **104** in the open position.

[0028] FIG. 2D illustrates a cutaway view of an exemplary key ring embodiment having an integral latch in the open position. To move from the closed to the open position the latch **104** is withdrawn into the second channel **112** which opens the gap **108** between the first end **124** and the second end **126** of the ring portion **102**. To move back to the closed position the latch **104** is moved outward from the second channel **112** to close the gap **108** and engage the first channel **122** and the catch **120** of the first end **124** of the ring portion **102**. In the closed position, the latch **104** is disposed within both the first channel **122** of the first end **124** and the second channel **112** of the second end **126**.

[0029] In a typical embodiment of the invention, the ring portion is bounded within a circular envelope and the latch **104** includes a contact area **110** which extends outside an outer radius of the circular envelope. The contact area can be a grooved surface or any other suitable pattern on a surface which can be pushed against by the finger of a user to facilitate movement of the latch **104** between the open and closed positions.

[0030] In some embodiments of the invention, distal from the second end **126** of the ring portion **102** an open stop **128** can be formed in the ring portion **102** as an edge of the ring portion **102**. In this case, the latch **104** further comprises an open stop latch **130** for engaging the open stop **128** when the latch **104** is in the open position. The open stop latch **130** can be formed at the back edge of the contact area **110**. This is similar to the opposing close stop **114** which is also formed in the ring portion **102** as an opposing edge of the ring portion **102** behind the open stop **128** more distal from the second end **126** of the ring portion **102**.

[0031] To make assembly of the key ring **100** convenient, the ring portion can include an opening **116** to the second channel **112** in an outer circumference of the ring portion **102** distal from the gap for receiving the latch **104** during assembly. The front end of the integral latch **104** is inserted into this opening **116** and pushed through the second channel **112** until it reaches the gap **108**. The pressure from the cantilever spring of the close stop latch **118** provides friction between the inner radial surface **134** of the second channel **112** and the back radial surface of the close stop **114** and the open stop **128**. This friction retains the latch **104** within the ring portion even in the open position when the first cantilever spring **106** is not engaged with the catch **120**.

[0032] The various embodiments of the invention described can be produced from any suitable resilient, durable materials. The key ring can be produced from metals, plastics or polymers. The material should be hard but also with sufficient spring to support engaging and disengaging of the catch and loop. Some example suitable metals include titanium or beryllium copper. However, casting, powder injection molding is also possible with titanium, stainless steel or other suitable metals or plastics. Composite materials can also be employed. For example, a steel or titanium core wrapped with a polycarbonate, e.g. a polycarbonate (or polymer) injected over a metal core. This type of construction will enable colorful designs in the polycarbonate or polymer.

[0033] This concludes the description including the preferred embodiments of the present invention. The foregoing description including the preferred embodiment of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible within the scope of the foregoing teachings. Additional variations of the present invention can be devised without departing from the inventive concept as set forth in the following claims.

Claims

1. An apparatus, comprising: a ring portion having a first end including a first channel and a catch and a second end including a second channel and a gap between the first end and the second end; and a latch disposed within the second channel of the second end for opening and closing the gap;

- wherein the latch is moveable between an open position such that the latch is withdrawn into the second channel opening the gap between the first end and the second end of the ring portion and a closed position such that the latch is moved outward from the second channel to close the gap and engage the first channel and the catch of the first end of the ring portion.
2. The apparatus of claim 1, wherein the ring portion is bounded within a circular envelope and the latch includes a contact area which extends outside the circular envelope.
 3. The apparatus of claim 1, wherein the latch comprises a cantilever spring for engaging the catch of the first end of the ring portion when the latch is in the closed position.
 4. The apparatus of claim 3, wherein the cantilever spring comprises a latch end turned back on itself.
 5. The apparatus of claim 1, wherein the latch further comprises an open stop latch and the ring portion further comprises an open stop and the open stop latch engages the open stop in the open position.
 6. The apparatus of claim 1, wherein the latch further comprises a closed stop latch and the ring portion further comprises a closed stop and the closed stop latch engages the closed stop in the closed position.
 7. The apparatus of claim 6, wherein the closed stop latch comprises a second cantilever spring.
 8. The apparatus of claim 7, wherein the second cantilever spring of the closed stop latch comprises a slotted back end of the latch.
 9. The apparatus of claim 1, wherein the ring portion further comprises an opening in an outer circumference of the ring portion to the second channel for receiving the latch during assembly.
 10. The apparatus of claim 1, wherein at least the ring portion comprises titanium.
 11. The apparatus of claim 1, wherein the latch and the ring portion comprise castings.
 12. The apparatus of claim 1, wherein the ring portion and the latch are cut using wire electro discharge machining (EDM).
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